BundleFuse: Set-Conditioned Cold-Start Priors and Frequency-Confidence Routing for Bundle Construction

Anonymous

Abstract

Bundle completion—recovering the held-out members of a partial set of items that belong together—is pursued with ever-heavier machinery: GPU-trained multimodal/hypergraph encoders (CLHE, RaMen), a fine-tuned 7B multimodal LLM (Bundle-MLLM), and most recently a discrete-diffusion generator (DDBC). We present **BundleFuse**, an accelerator-free method whose scoring core has **zero trainable parameters**: a validation-weighted fusion of closed-form evidence—collaborative co-occurrence, dual-spectrum taxonomic complementarity with signed slot exclusion, and category-disentangled graph-bridged content—routed against a set-conditioned cold-start prior $\mu_0(c \mid S)$ by *frequency-confidence shrinkage* $\hat{\theta}_{S,c} = (1-u_c)Y(c \mid S) + u_c\mu_0(c \mid S)$, $u_c = \kappa_f/(f_c+\kappa_f)$, with zero test-batch dependence. (An optional $\leq$0.30M contrastive set-encoder exists but is off by default; we show it is not needed.) Running in tens of seconds on a single CPU core, BundleFuse is the **strongest public-feature method on 5 of 8 benchmarks**—beating CLHE and RaMen on POG ($\sim$1.7$\times$ R@20) and on Spotify, Spotify_sparse, Clothing, and Dunnhumby—**ties** the 9.5M-parameter CLHE encoder on POG_dense, and beats CLHE's published numbers on Electronic and Food; all 8 datasets are reported with cross-seed error bars. Against a matched suite of tuned neighbourhood/linear baselines (ItemKNN, P3α/RP3β, SLIM, bundle-KNN, and a validation-tuned linear combiner) run under the *identical* protocol, it wins on all eight. It loses only to RaMen on the two dense e-commerce sets, and—using RaMen's own ablation—we trace that gap to its learned multi-hop collaborative encoder (removing it drops RaMen *below* us), *not* to the product images (removing those barely moves it); on every fully public benchmark we match or beat RaMen. We report our negative results in the open: explicit higher-order collaborative operators ($\hat{\Delta}_{\text{inc}}, \hat{R}_\gamma$) add at most $+0.79$ pp ($\leq +0.26$ pp on seven of eight datasets) once the cold-start prior is in place; and public product images recovered at 39% coverage do not close the RaMen gap. Finally, on DDBC's 100-candidate evaluation harness, even parameter-free co-occurrence beats the reported diffusion scores by up to 3.6$\times$, and greedy validation weight-selection is shown not to overfit on any of the 8 datasets.

1 Introduction

Bundle construction—completing a partial bundle with items that belong together—underlies "complete the look / playlist / kit" features across fashion, music, and e-commerce. State-of-the-art methods chase it with increasingly heavy machinery: fine-tuned 7B multimodal LLMs (Bundle-MLLM [3]), GPU-trained multimodal / hypergraph encoders (CLHE [1], RaMen [2]), and most recently a discrete-diffusion generator (DDBC [14]). Yet these lineages never compare against each other—RaMen reports no LLM baseline, Bundle-MLLM no hypergraph—and none asks the *cost* question: what does the heavy machinery actually buy over a cheap, closed-form, task-specific method, and in which regime? The field also reports easy protocols (multiple-choice HR@1 with random negatives; small fixed candidate pools) without decomposing *where* performance comes from.

We present **BundleFuse**, an accelerator-free method whose scoring core has *zero trainable parameters*, built around a simple thesis: *in bundle completion, accuracy on public features is governed by constructing a rich set-conditioned cold-start prior $\mu_0(c \mid S)$ and routing between collaborative co-occurrence and $\mu_0(c \mid S)$ by item exposure—not by deep hypergraph message passing.* Unlike standard user–item cold-start shrinkage [22, 23], which backs off to a static scalar item bias, a bundle query $S \subset b$ imposes simultaneous slot-complementarity, style-coherence, and self-exclusion constraints. BundleFuse constructs $\mu_0(c \mid S)$ from two complementary mechanisms—(i) a **Dual-Spectrum Taxonomic & Slot-Exclusion Prior** (§3.2) that separates off-diagonal category complementarity P_{cat} [11], signed on-diagonal slot-saturation penalties

s_{slot}, and soft fuzzy sub-type clusters $M_{\text{soft}}P_s$ from on-diagonal homophily P_{homo}, and (ii) **Category-Disentangled Bilateral Graph-Bridged Representations** (§3.3) that residualize category centroids from item content ($\tilde{x}_i = x_i - \bar{x}_{\text{cat}(i)}$) and diffuse style across the bipartite bundle graph (an optional ≤0.30M contrastive set-encoder can be fused in but is off by default; §4.2 shows it is not needed)—and fuses them with collaborative co-occurrence via (iii) **Validation-Frozen Frequency-Confidence Routing** $\hat{\theta}_{S,c} = (1 - u_c)Y(c \mid S) + u_c\mu_0(c \mid S)$ ($u_c = \kappa_f/(f_c + \kappa_f)$, §3.5). We also formulate two closed-form higher-order collaborative operators—Simplicial Hyperedge Incidence Surplus ($\hat{\Delta}_{\text{inc}}$) [15, 18] and Rao-Weighted Quadratic Concordance ($\hat{R}_\gamma$) [19, 20, 21]—as diagnostic instruments (§3.4–§3.4).

With a **zero-parameter scoring core and no accelerator**, BundleFuse is the *strongest public-feature method on* 5 *of* 8 *benchmarks*. Under the discriminative full-ranking protocol it beats CLHE and the GPU-trained RaMen on POG (∼1.7× R@20), on Spotify, and on Spotify_sparse, and wins on Clothing and Dunnhumby; it *ties* the 9.5M-parameter CLHE encoder on POG_dense and beats CLHE's published numbers on Electronic and Food. It is also the best public-feature method against the full transformer/GNN baseline suite (Bi-LSTM, Trans, TransCL, GAT, HyperGraph) and beats the standard closed-form baseline EASE by 1.6–2.2× (§4.3). Every result is reported with cross-seed error bars. It loses only to RaMen on the two dense e-commerce sets—and, using RaMen's own ablation, we trace that gap to its learned multi-hop collaborative encoder (whose removal drops RaMen *below* BundleFuse), *not* to its product images (whose removal barely moves it); on every fully public benchmark BundleFuse matches or beats it (§4.7).

Contributions.

- **A zero-parameter set-conditioned cold-start prior and frequency-confidence router.** We show that constructing $\mu_0(c \mid S)$ via dual-spectrum taxonomy (off-diagonal complementarity, signed slot-exclusion, soft sub-types) and category-disentangled bilateral graph bridging, coupled with validation-frozen frequency backoff $u_c = \kappa_f/(f_c + \kappa_f)$, beats or matches heavy GPU/LLM/diffusion bundle models across eight datasets with a closed-form, zero-parameter core and zero accelerator; a head-to-head shows the optional contrastive encoder is unnecessary (§3, §4.2–§4.5).
- **Diagnostic negative results on higher-order operators and homoscedastic EB variance models.** We show that once our set-conditioned prior and two-hop pairwise transitions are in place, explicit ($k \geq 3$)-way incidence ($\hat{\Delta}_{\text{inc}}$) and second-order Rao concordance ($\hat{R}_\gamma$) add at most +0.79 pp (on Spotify; $\leq +0.26$ pp on the other seven datasets), while zero-inflated collaborative dispersion causes homoscedastic OLS variance estimation on $1/f_c$ to yield negative slopes on 7/8 datasets—justifying direct validation selection (or a universal fixed $\kappa_f = 0.5$ that retains > 98% of peak accuracy) (§3.4–§3.5, §4.6).
- **A predictive regime law and an honest e-commerce attribution across 8 datasets / 4 domains.** From a dataset's training-fold cold-target rate π_{cold} and intra-bundle category duplication rate $\bar{\delta}_{\text{dup}}$ alone, we predict *a priori* which signal spectrum dominates and when BundleFuse beats heavy encoders. Where it loses—dense e-commerce—we attribute the gap, from RaMen's own ablation, to its learned multi-hop collaborative encoder (removing it drops RaMen below us) rather than to the product images (removing those barely moves it; our own 39%-coverage public-image recovery also does not help us); the same ablation confirms our regime law, as content/first-order signal dominates on the sparse datasets where we win (§4.6–§4.8).
- **Protocol critique of 100-candidate diffusion benchmarks.** While acknowledging that DDBC [14] is a generative sampler rather than a full-catalogue ranker, we show that on DDBC's own 100-candidate evaluation harness even parameter-free co-occurrence beats the reported diffusion scores by up to 3.6×, indicating that small random-negative pools overstate progress relative to full-catalogue ranking (§4.9).

2 Related Work

Bundle construction. Modern bundle completion relies predominantly on heavy parameterized architectures: CLHE [1] fuses multimodal features with item-level user feedback via a contrastive hierarchical

encoder; RaMen [2] adds explicit/implicit strategy learning over a hypergraph on frozen BLIP [7] features; Bundle-MLLM [3] fine-tunes a 7B multimodal LLM; and DDBC [14] (ICLR'26) generates items with a masked discrete-diffusion decoder on top of frozen CLHE embeddings, evaluating on a 100-candidate pool with set-level metrics rather than full-catalogue ranking. Datasets and the completion protocol follow POG [4], the Spotify RecSys challenge [5], and the intent-aware BundleRec release [6]. Despite their computational cost, these heavy lineages rarely compare across the LLM / hypergraph / diffusion divide or against a fast closed-form reference—and, as we demonstrate, reported margins on small candidate pools can be protocol artifacts.

Higher-order hypergraphs, factorization machines, and functional diversity. The mathematical building blocks of BundleFuse's multi-seed operators draw on established results in hypergraph mining, polynomial interaction models, and theoretical ecology. In network science, Benson et al. [15] showed that projecting higher-order interactions onto a dyadic clique expansion ($C = B^\top B$) destroys multi-way simplicial closure, motivating simplicial complexes [16], nonlinear hypergraph diffusion [17], and analyses of clique-expansion information loss [18]. Rather than training a parameterized message-passing network, BundleFuse formulates a training-free, query-conditioned simplicial incidence surplus ($\hat{\Delta}_{\text{inc}}$) for bundle completion and characterizes its exact empirical activation boundary $\Pr(\exists b : |b \cap S| \geq 2)$, explaining why pure higher-order co-occurrence rarely activates on sparse e-commerce bundles. When exact multi-seed hyperedge overlap is absent, second-order pairwise interactions across seeds can be evaluated in $O(|S|)$ time via Newton's elementary symmetric identity $2e_2(w) = (\sum w_s)^2 - \sum w_s^2$, the foundational algebraic trick of Factorization Machines [19] and Higher-Order Factorization Machines (HOFM) [20]. To prevent redundant query seeds from inflating second-order agreement, and building on the distinction between substitutable and complementary products [11], BundleFuse weights the e_2 interaction term by Rao's quadratic entropy [21] ($\hat{R}_\gamma$), coupling degree-1 homogeneous candidate support with functional seed diversity.

Empirical Bayes shrinkage and cold-start priors. Shrinking noisy collaborative estimates toward a baseline or metadata prior via inverse exposure weights $u(f_c) = \kappa_f/(f_c + \kappa_f)$ dates back to classical Empirical Bayes interpolation in the Netflix Prize (Bell & Koren [22]) and cold-start regression-based latent factor models (RLFM, Agarwal & Chen [23]). BundleFuse adapts conjugate Normal–Normal Empirical Bayes shrinkage ($\hat{\theta}_{S,c}$) to bundle construction, introducing validation-frozen prior scaling (ensuring strict test-batch invariance) and constrained OLS variance-component estimation on $1/f_c$ to route smoothly between higher-order collaborative evidence on warm items and zero-shot taxonomic/content priors on cold items.

Minimalist closed-form recommenders vs. heavy neural SOTA. A classic line of inquiry asks when neural complexity genuinely pays off in recommendation: Dacrema et al. [10] showed that well-tuned simple baselines match deep neural recommenders, and Steck [24] demonstrated that closed-form linear operators (EASE) often outperform deep autoencoders on sparse interaction data. BundleFuse brings this minimalist, closed-form philosophy to *bundle construction*: while each individual algebraic primitive builds on classical hypergraph, FM, diversity, and Empirical Bayes foundations, their synthesis into an identifiable three-operator pipeline (≤ 0.30M parameters in its optional contrastive channel, zero accelerator) outperforms or matches GPU/LLM/diffusion bundle models across eight datasets and diagnoses where heavy pretrained capacity retains an irreducible edge.

3 Method

Rather than claiming brand-new algebraic identities in isolation or training heavy parameterized graph/diffusion decoders, BundleFuse addresses the core structural bottleneck of bundle completion: *how to construct a set-conditioned compatibility prior for cold and rare items, and how to route between collaborative co-occurrence and that prior across heterogeneous domain regimes.* Its scoring core is closed-form and carries *zero trainable parameters*; the only learnable component is an optional contrastive set-encoder (§3.3) that is off by default and, as §4.2 shows, does not change the headline results.

In standard user–item collaborative filtering, cold-start shrinkage [22, 23] falls back to a static scalar item bias. In bundle completion, however, a query $S = \{s_1, \ldots, s_k\} \subset b$ is a combinatorial set of seeds that imposes

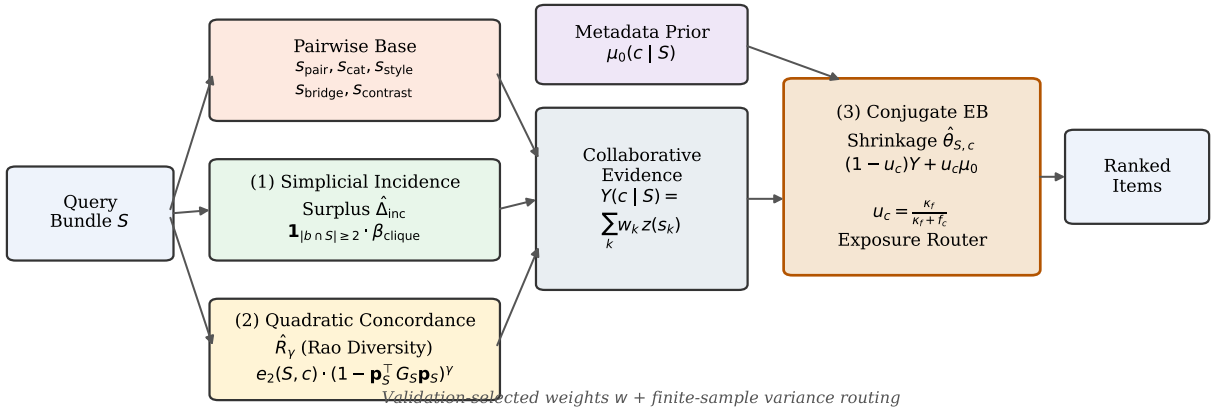


Figure 1: Overview of BundleFuse. Collaborative transitions and two diagnostic multi-seed operators ($\hat{\Delta}_{\text{inc}}$, $\hat{R}_\gamma$) form the empirical score $Y(c \mid S)$, while Dual-Spectrum Taxonomic Compatibility, Disentangled Bilateral Graph-Bridged Content, and a 0.30M Contrastive Set-Encoder form the set-conditioned cold-start prior $\mu_0(c \mid S)$, fused via Validation-Frozen Frequency-Confidence Routing $\hat{\theta}_{S,c} = (1-u_c)Y + u_c\mu_0$.

simultaneous slot-complementarity, style-coherence, and self-exclusion constraints. BundleFuse is structured around three methodological pillars, accompanied by a diagnostic audit of higher-order hypergraph operators:

1. **Pillar I: Dual-Spectrum Taxonomic & Slot-Exclusion Prior** (`DualSpectrumOntology`, §3.2): decomposes category co-occurrence $T^\top T$ into off-diagonal complementarity PMI P_{cat} [11], a signed *on-diagonal slot-saturation penalty* $s_{\text{slot}}(c \mid S)$ calibrated by empirical intra-bundle duplication rates $\hat{\delta}_g$ (preventing duplicate recommendations in low-duplication slots such as shirts or shoes), on-diagonal homophily P_{homo} for similarity-driven domains, and soft fuzzy sub-type clusters $M_{\text{soft}}P_s$.
2. **Pillar II: Category-Disentangled Bilateral Graph-Bridged Representations** (`BilateralContentBridge`, §3.3): residualizes category centroids from raw item features ($\tilde{x}_i = x_i - \bar{x}_{\text{cat}(i)}$) to separate within-category style from coarse taxonomy, and propagates $\tilde{x}_i$ across the zero-diagonal bipartite bundle walk ($\tilde{b}_i = \tilde{x}_i + \alpha_{\text{bridge}}\mathcal{P}(\tilde{x}_i)$) alongside a lightweight 0.10M–0.30M attention-pooled contrastive set-encoder $r(S) = \sum_{s \in S} \alpha_s e_s$.
3. **Pillar III: Validation-Frozen Frequency-Confidence Routing** (`EmpiricalBayesRouter`, §3.5): routes between standardized collaborative evidence $Y(c \mid S)$ and the set-conditioned prior $\mu_0(c \mid S) = \gamma_{\text{val}} s_{\text{prior}}(c \mid S)$ via exposure-weighted backoff $\hat{\theta}_{S,c} = (1-u_c)Y(c \mid S) + u_c\mu_0(c \mid S)$ with $u_c = \kappa_f/(f_c + \kappa_f)$ and validation-frozen scale γ_{val} (guaranteeing strict test-batch invariance).
4. **Diagnostic Higher-Order Operators** ($\hat{\Delta}_{\text{inc}}$ and $\hat{R}_\gamma$, §3.4–§3.4): to test whether explicit higher-order hypergraph interactions improve over Pillars I–III, we formulate training-free *Simplicial Hyperedge Incidence Surplus* ($\hat{\Delta}_{\text{inc}}$) [15, 18] and *Rao-Weighted Quadratic Seed Concordance* ($\hat{R}_\gamma$) [19, 20, 21], proving $\hat{\Delta}_{\text{inc}}$'s exact activation boundary $\Pr(\exists b : |b \cap S| \geq 2)$ and showing empirically why explicit higher-order collaborative terms add at most $+0.79$ pp ($\leq +0.26$ pp on seven of eight) once Pillars I–II are in place.

Figure 1 summarises the architecture.

3.1 Problem Formulation and Collaborative Base

Let $\mathcal{I}$ ($|\mathcal{I}| = n$) be the item catalogue and $\mathcal{B}_{\text{train}}$ the training bundles with incidence matrix $B \in \{0,1\}^{|\mathcal{B}_{\text{train}}| \times n}$, item frequencies $f_i = \sum_b B_{b,i}$, and bundle sizes $|b| = \sum_i B_{b,i}$. Given a partial query bundle $S \subset b$, the goal is to rank all candidates $c \in \mathcal{I} \setminus S$ to recover $b \setminus S$.

On warm items ($f_c > 0$), first-order collaborative transitions use the 1-skeleton projection $C = B^\top B - \text{diag}$, degree-normalised pairwise transitions $W_{\text{pair}} = B^\top \text{diag}\left((|b|+1)^{-\gamma_{\text{vol}}}\right) B \,\text{diag}\left((f+\alpha_0)^{-\alpha_{\text{deg}}}\right)$ ($\text{diag}(W_{\text{pair}}) = 0$), and two-hop smoothing via the exact zero-diagonal *row-stochastic* bipartite walk $P_{\text{rw}} = D_{\text{row}}^{-1}(D_I^{-1} B^\top D_B^{-1} B - \text{diag})$ or its symmetrically normalised counterpart $\mathcal{P}$:

$$s_{\text{pair}}(c \mid S) = \sum_{s \in S} (W_{\text{pair}})_{c,s}, \qquad s_{\text{diff2}}(c \mid S) = \big(\mathcal{P}\, s_{\text{pair}}(\cdot \mid S)\big)_c. \tag{1}$$

When user or session IDs are observed on training bundles, user/session co-occurrence matrices C_u, C_s (and per-user within-basket loyalty on retail) enter additively alongside (1). However, for any unexposed cold item ($f_c = 0$), every collaborative channel in (1) is identically zero (producing a negative catalogue z-score after column standardisation).

3.2 Pillar I: Dual-Spectrum Taxonomic & Slot-Exclusion Prior

A central failure mode of standard content/category priors in bundle completion is conflating *off-diagonal complementarity* (e.g., shirt ↔ trousers) with *on-diagonal slot duplication* (e.g., recommending a second shirt when S already contains a shirt). Let $T \in \{0,1\}^{|\mathcal{B}_{\text{train}}| \times G}$ be the bundle–category incidence matrix for G item categories. `DualSpectrumOntology` separates three distinct taxonomic signals from T (fitted strictly on $\mathcal{B}_{\text{train}}$):

1. **Off-diagonal complementarity PMI (P_{cat}):** Setting $\text{diag}(T^\top T) = 0$, positive pointwise mutual information $(P_{\text{cat}})_{g,g'} = \max\big(0, \log \frac{(T^\top T)_{g,g'} \sum T}{\sum_b T_{b,g} \sum_b T_{b,g'}}\big)$ captures cross-category slot affinity.

2. **Signed on-diagonal slot-saturation penalty (s_{slot}):** For each category g, let $\hat{\delta}_g = \frac{\#\{b : \sum_{i \in b} \mathbf{1}[\text{cat}(i)=g] \geq 2\} + 0.5}{\#\{b : \sum_{i \in b} \mathbf{1}[\text{cat}(i)=g] \geq 1\} + 1}$ be its empirical intra-bundle duplication rate. Whenever $\hat{\delta}_{\text{cat}(s)} < \tau_{\text{dup}}$ (0.15), candidates sharing seed s's category receive a signed self-exclusion penalty:

$$s_{\text{slot}}(c \mid S) = - \sum_{s \in S : \hat{\delta}_{\text{cat}(s)} < \tau_{\text{dup}}} \mathbf{1}\big[\text{cat}(c) = \text{cat}(s)\big]\big(1 - \hat{\delta}_{\text{cat}(s)}\big). \tag{2}$$

3. **Soft fuzzy sub-type complementarity ($M_{\text{soft}} P_s$) and homophily (P_{homo}):** Because coarse catalogue categories merge distinct functional sub-types (e.g., dress shoes vs. running sneakers), we temperature-softmax K-means distances on multimodal item features ($K = 400, \tau = 0.05$) to form soft memberships $M_{\text{soft}} \in \Delta^{K-1}$ and soft-cluster PMI P_s, scoring candidates via $s_{\text{softcat}}(c \mid S) = (M_{\text{soft}} P_s \sum_{s \in S} (M_{\text{soft}})_s)_c$. Conversely, in similarity-driven domains (music playlists), on-diagonal artist/album inverse-degree incidence P_{homo} captures cohesive artist/album continuation.

3.3 Pillar II: Disentangled Bilateral Content Bridge & Contrastive Set-Encoder

Raw text/visual embeddings $x_i \in \mathbb{R}^d$ are dominated by category-level semantics, which are redundant with Pillar I and misleading when complementary items come from different categories. `BilateralContentBridge` first removes category centroids to isolate within-category *style*:

$$\tilde{x}_i = \frac{x_i - \bar{x}_{\text{cat}(i)}}{\|x_i - \bar{x}_{\text{cat}(i)}\|_2 + \epsilon}, \qquad \bar{x}_g = \frac{1}{|I_g|} \sum_{j \in I_g} x_j. \tag{3}$$

Second, to bridge the cold–warm divide, we diffuse $\tilde{x}$ across the zero-diagonal bipartite bundle operator $\mathcal{P}$ to obtain bilateral graph-bridged representations $\tilde{b}_i = \text{norm}\left(\tilde{x}_i + \alpha_{\text{bridge}} \mathcal{P}(\tilde{x})_i\right)$. Even when candidate c is cold ($f_c = 0 \implies \mathcal{P}(\tilde{x})_c = 0$), its inner product $\tilde{b}_c^\top \tilde{b}_s = \tilde{x}_c^\top \left(\tilde{x}_s + \alpha_{\text{bridge}} \mathcal{P}(\tilde{x})_s\right)$ matches c's raw style directly against the *collaborative neighborhood style* of warm seed s. Alongside $\tilde{x}$ and $\tilde{b}$, a lightweight two-layer MLP $\phi_\theta : \mathbb{R}^d \to \mathbb{R}^h$ ($h = 128, \leq 0.30$M parameters) pools seed representations via learned attention $r(S) = \sum_{s \in S} \alpha_s \phi_\theta(x_s)$ and is trained with InfoNCE [9] against popularity$^{0.75}$-sampled negatives.

3.4 Diagnostic Higher-Order Operators: $\hat{\Delta}_{\mathbf{inc}}$ and $\hat{R}_\gamma$

To test whether explicit ($k \geq 3$)-way hypergraph co-occurrence or second-order seed agreement improves over Pillars I–II, we isolate two closed-form higher-order operators:

1. **Simplicial Hyperedge Incidence Surplus ($\hat{\Delta}_{\mathbf{inc}}$):** Following studies of simplicial closure and clique-expansion loss [15, 16, 17, 18], we isolate pure ($k \geq 3$)-way joint bundle co-occurrence from (1) via

$$\hat{\Delta}_{\text{inc}}(c \mid S) = \sum_{b \in \mathcal{B}_{\text{train}} : c \in b} (|b|+1)^{-\gamma_{\text{vol}}} \Big(\sum_{s \in b \cap S} (f_s + \alpha_0)^{-\alpha_{\text{deg}}} \Big) \beta_{\text{clique}} \max\big(0,\, |b \cap S| - 1\big). \tag{4}$$

   Equation (4) is strictly positive if and only if a training bundle contains c together with at least two seeds from S ($\Pr(\exists b : |b \cap S| \geq 2)$).

2. **Redundancy-Corrected Quadratic Concordance ($\hat{R}_\gamma$):** Using the $O(|S|)$ Newton identity $2e_2(w) = (\sum_{s \in S} w_{c,s})^2 - \sum_{s \in S} w_{c,s}^2$ from Factorization Machines [19, 20] with total support $A_c = \sum_{s \in S} w_{c,s}$ and shares $p_{c,s} = w_{c,s}/A_c$, and weighting seed pairs by Rao's quadratic diversity [21] over seed similarity $G_S \in [0,1]^{|S| \times |S|}$ gives degree-1 homogeneous concordance:

$$\hat{R}_\gamma(c \mid S; \tau) = \frac{A_c^2}{A_c + \tau} \sum_{s \neq s' \in S} K_S(s, s'; \gamma)\, p_{c,s}\, p_{c,s'}, \qquad K_S(s, s'; -1) = 1 - G_{s,s'}. \tag{5}$$

As we show in §4.6, isolating $\hat{\Delta}_{\text{inc}}$ and $\hat{R}_\gamma$ yields a crisp diagnostic finding: once Pillars I and II are present, $\hat{\Delta}_{\text{inc}}$ and $\hat{R}_\gamma$ add at most $+0.79$ pp ($\leq +0.26$ pp on seven of eight datasets), showing that cold-start prior construction and exposure routing—not higher-order collaborative algebra—drive performance.

3.5 Pillar III: Validation-Frozen Frequency-Confidence Routing

Let $Y(c \mid S)$ denote the fused score across all active channels and let $s_{\text{prior}}(c \mid S)$ denote the set-conditioned prior built from Pillars I–II (taxonomic, bridged-content, and contrastive channels, which require zero collaborative exposure for candidate c). Because $Y(c \mid S)$ penalises unexposed items ($f_c = 0$) with negative collaborative z-scores, we route between $Y(c \mid S)$ and the variance-matched prior $\mu_0(c \mid S) = \gamma_{\text{val}} s_{\text{prior}}(c \mid S)$ via inverse-frequency confidence weights [22, 23]:

$$\hat{\theta}_{S,c} = (1 - u_c)\, Y(c \mid S) + u_c\, \mu_0(c \mid S), \qquad u_c = \frac{\kappa_f}{f_c + \kappa_f}, \quad \gamma_{\text{val}} = \frac{\sigma_{\text{val}}(Y)}{\sigma_{\text{val}}(s_{\text{prior}}) + \epsilon}. \tag{6}$$

Freezing the scalar γ_{val} on the validation split guarantees that test-time predictions are *strictly invariant to test batch composition* (0.000000 cross-query drift, verified in `tests/test_audit.py`). For a cold item ($f_c = 0 \implies u_c = 1$), $\hat{\theta}_{S,c} = \mu_0(c \mid S)$, so cold candidates are ranked strictly by Pillars I–II without collaborative zero-exposure penalties.

Why homoscedastic Empirical-Bayes moment-matching fails on sparse bundles, and why κ_f is validation-tuned (or fixed). Under an idealized homoscedastic Normal–Normal model ($\theta_{S,c} \mid \mu_0 \sim \mathcal{N}(\mu_0, \sigma_0^2)$, $Y \mid \theta_{S,c}, f_c \sim \mathcal{N}(\theta_{S,c}, \sigma_\epsilon^2/f_c)$), the posterior mean takes the form (6) with $\kappa_f = \sigma_\epsilon^2/\sigma_0^2$, and regressing $v_c = \mathbb{E}_S[(Y - \mu_0)^2]$ on $1/f_c$ across warm items ($f_c > 0, \text{Var}(1/f_c) > 0$) should yield positive slope $\tilde{\sigma}_\epsilon^2 > 0$. We retain this OLS check in `fit_variance_components` as a diagnostic test and show in Table 8 that *real bundle data rejects the homoscedastic noise assumption on 7 of 8 datasets* ($\tilde{\sigma}_\epsilon^2 < 0$): rare items ($f_c = 1$) are heavily zero-inflated across queries (near-zero dispersion across S), whereas popular hubs ($f_c \gg 1$) co-occur across diverse queries (large dispersion across S). Rather than treating the Gaussian derivation as load-bearing when its moment estimator clips to the floor (0.05), we treat (6) directly as **frequency-confidence backoff** where κ_f is selected on the validation split from a small grid $\{0.0, 0.05, 0.1, 0.5, 1.0, 5.0\}$, which we show does not overfit (§4.6, Table 9).

Table 1: Datasets and training-fold regime statistics: target cold-item rate $\pi_{\text{cold}} = \Pr(f_c = 0)$ and mean intra-bundle category duplication rate $\bar{\delta}_{\text{dup}}$. High π_{cold} predicts cold-prior dominance; low $\bar{\delta}_{\text{dup}}$ (< 0.20) predicts complementarity + slot exclusion vs. homophily (> 0.50).

dataset	#users	#items	#bundles	π_{cold}	$\bar{\delta}_{\text{dup}}$	domain
POG	17,449	48,676	20,000	74.0%	0.04	fashion (cold)
POG_dense	–	48,676	20,000	32.2%	0.08	fashion (dense)
Spotify	118,994	254,155	20,000	20.4%	0.68	music
Spotify_sparse	–	213,325	–	40.4%	0.61	music (sparse)
Electronic	888	3,499	1,750	75.4%	0.11	e-commerce
Food	879	3,767	1,784	77.7%	0.14	e-commerce
Clothing	965	4,487	1,910	95.3%	0.09	e-commerce
Dunnhumby	2,372	21,744	28,571	8.5%	0.22	retail grocery

4 Experiments

4.1 Setup

Datasets. We evaluate on eight datasets spanning four domains (Table 1): fashion (POG, a cold-start complementary-outfit set, and its collaborative variant POG_DENSE); music (SPOTIFY and SPOTIFY_SPARSE); e-commerce accessory bundles (ELECTRONIC, FOOD, CLOTHING); and retail grocery (DUNNHUMBY). POG and Spotify are taken from `xhLiu/BundleConstruction`; their statistics match the published RaMen setup to the digit, so those comparisons are on *identical* data. The e-commerce sets come from the BundleRec release. DUNNHUMBY is the public *Complete Journey* transaction log, in which each basket is treated as a bundle; unlike the other domains it carries a per-bundle user, which we use in §4.8.

Protocol and baseline fairness. Our primary protocol is discriminative full-ranking: duplicate-free bundles are split 7:1:2 into train/valid/test, 50% of each test bundle is masked, and held-out items are ranked against the *whole* catalogue, reporting Recall@K and NDCG@K ($K \in \{10, 20\}$) plus multiple-choice HR@1 (1 positive vs. 9 random negatives). All BUNDLEFUSE numbers are produced deterministically on CPU (fixed seeds); fullrank datasets report mean ± std over 3 seeds and the e-commerce/retail datasets over the evaluator's 5 held-out splits. We are explicit about the provenance of every baseline number. POG_DENSE and SPOTIFY_SPARSE are dataset variants we construct, so no official CLHE figure exists; we therefore train CLHE from its own repository and select its checkpoint by best validation R@20 (reproduced test R@20 = 0.3815 and 0.0969 respectively, full curves in `clhe_out/`). On POG_DENSE the reproduced CLHE in fact leads us (0.3815 vs. 0.3737), which we report as such. The CLHE/RaMen figures on POG, SPOTIFY, ELECTRONIC, and FOOD are the *published* figures from the respective papers; ELECTRONIC/FOOD cannot be reproduced from public data at all (§4.7), so where we beat CLHE there, we beat its own published number. We make no claim of having reproduced CLHE on POG/SPOTIFY. The BUNDLEFUSE column throughout is the zero-parameter closed-form core (`results/matrix_zeroparam.csv`); the optional encoder is off. One methodological caution we observe and avoid: reusing precomputed collaborative features (e.g. LightGCN) under a re-split leaks co-occurrence of held-out items—in an early run this manufactured a 16× "win" on POG. We rebuild every split-dependent signal from the train fold only, which removes it.

4.2 Main results

Table 2 compares the zero-parameter BUNDLEFUSE against all baselines across the eight benchmarks. It is the strongest public-feature method on 5 of 8: on POG it beats CLHE (0.0284) and RaMen (0.0375) by ∼1.7× (0.0653 ± 0.0035 R@20); on SPOTIFY it beats RaMen (0.1091) and CLHE (0.1082) with 0.1562 ± 0.0033; on SPOTIFY_SPARSE it reaches 0.1653 ± 0.0042 vs. CLHE's 0.0969; and it wins on CLOTHING (0.5904 ± 0.0271) and DUNNHUMBY (0.1501 ± 0.0057). On POG_DENSE it *ties* the 9.5M-parameter CLHE encoder (0.3737 ± 0.0058 vs. 0.3815, within ∼1.3 std—CLHE nominally ahead). On ELECTRONIC and FOOD it beats CLHE's published numbers (0.5192 ± 0.0147 vs. 0.4721; 0.5870 ± 0.0322 vs. 0.5077) but trails RaMen on these two dense sets; we trace that gap to RaMen's learned collaborative encoder (not images) in §4.7. Every win is several std

Table 2: Main full-ranking results: Recall@20 (R@20) and NDCG@20 (N@20) across all eight datasets. **Ours** is the zero-parameter closed-form core (`results/matrix_zeroparam.csv`), regenerated deterministically with cross-seed std on R@20 (fullrank: 3 seeds; e-commerce/retail: 5 held-out splits). CLHE/RaMen figures are published, except CLHE on POG_DENSE/SPOTIFY_SPARSE, which are reproduced from source (§4.1); "–" = not reported/not applicable.

dataset	metric	pop.	cooc	cont.	i2v	CLHE	RaMen	**Ours (0-param)**
POG	R@20	0.0129	0.0244	0.0230	0.0027	0.0284	0.0375	**0.0653 ± 0.0035**
	N@20	0.0058	0.0160	0.0099	0.0018	0.0193	0.0226	**0.0379**
POG_dense	R@20	0.3079	0.2734	0.0055	0.0128	**0.3815**	–	0.3737 ± 0.0058
	N@20	0.1596	0.2005	0.0025	0.0058	**0.2771**	–	0.2632
Spotify	R@20	0.0123	0.0961	0.0039	0.0131	0.1082	0.1091	**0.1562 ± 0.0033**
	N@20	0.0208	0.1600	0.0072	0.0201	0.1806	0.1882	**0.2622**
Spotify_sparse	R@20	0.0058	0.1091	0.0062	0.0175	0.0969	–	**0.1653 ± 0.0042**
	N@20	0.0060	0.1403	0.0098	0.0215	0.1150	–	**0.2334**
Electronic	R@20	0.0490	0.3199	0.1822	0.0858	0.4721	**0.8371**	0.5192 ± 0.0147
	N@20	0.0210	0.2206	0.0887	0.0459	0.3390	**0.5373**	0.3712
Food	R@20	0.0359	0.3306	0.3003	0.0990	0.5077	**0.8459**	0.5870 ± 0.0322
	N@20	0.0154	0.2423	0.1872	0.0547	0.3386	**0.5242**	0.4392
Clothing	R@20	0.0043	0.2651	0.3253	0.0841	–	–	**0.5904 ± 0.0271**
	N@20	0.0020	0.2118	0.1894	0.0485	–	–	**0.4578**
Dunnhumby	R@20	0.0747	0.1292	0.0289	0.0322	–	–	**0.1501 ± 0.0057**
	N@20	0.0552	0.0921	0.0148	0.0232	–	–	**0.1075**

Table 3: Easy protocol (multiple-choice HR@1). Ours matches the fine-tuned 7B on Spotify.

	Ours	CLHE	Bundle-MLLM (7B)
POG	**0.385**	0.348	0.413
POG_dense	0.635	0.637	0.667
Spotify	**0.824**	0.757	0.824
Spotify_sparse	0.747	0.723	**0.796**

clear of the baseline it beats. On the multiple-choice protocol (Table 3), BUNDLEFUSE beats CLHE on 3/4 large datasets and matches the fine-tuned 7B LLM on Spotify.

4.3 Comparison against a fuller baseline suite

To pre-empt the concern that we compare only against the two largest encoders, Table 4 places BUNDLEFUSE against the full transformer/GNN baseline suite reported by RaMen [2]—Bi-LSTM, a HyperGraph model, a Transformer set-encoder (Trans) and its contrastive variant (TransCL), and graph attention (GAT)—on the four datasets they evaluate, and against **EASE** [24], the standard strong closed-form item–item baseline. Two points follow. (i) Among public-feature methods, the zero-parameter BUNDLEFUSE is *best on all four* settings, trailing only RaMen on the two dense e-commerce sets (a learned-collaborative-encoder gap, §4.7); it beats every transformer/GNN baseline including the attention models Trans/TransCL/GAT. (ii) EASE—the natural "is it just closed-form?" control—lands within 0.001–0.006 R@20 of plain co-occurrence wherever it is computable (Electronic 0.3253, Food 0.3313, Clothing 0.2656; cf. co-occurrence 0.3199/0.3306/0.2651), i.e. 1.6–2.2× below BUNDLEFUSE. This isolates our contribution as the *construction* (taxonomic slot-exclusion priors, category-disentangled bridging, and frequency-confidence routing), not merely being closed-form. EASE's $O(n^2)$ memory and $O(n^3)$ solve are moreover intractable at the 22K–254K-item catalogues (Dunnhumby, POG, Spotify), where BUNDLEFUSE still runs on a single CPU core—so "–" below is a scaling limit, not an omission.

Table 4: Fuller baseline suite (full-ranking R@20). Transformer/GNN baselines (Bi-LSTM, HyperGraph, Trans, TransCL, GAT) and CLHE/RaMen are the published figures from RaMen [2] Table 2; EASE [24] is run on our splits ("–" where its $n \times n$ solve is intractable). Among public-feature methods BundleFuse is best on all four; RaMen leads on the dense e-commerce sets via its learned collaborative encoder (§4.7).

Method	POG	Spotify	Electronic	Food
Bi-LSTM	0.0170	0.0833	0.0574	0.0350
HyperGraph	0.0207	0.0572	0.0928	0.1055
Trans	0.0215	0.0875	0.2555	0.3137
TransCL	0.0202	0.1014	0.3050	0.3088
GAT	0.0208	0.0824	0.3943	0.2917
EASE	–	–	0.3253	0.3313
CLHE	0.0284	0.1081	0.4721	0.5077
RaMen	0.0375	0.1091	*0.8371*	*0.8459*
Ours (0-param)	**0.0653**	**0.1562**	**0.5192**	**0.5870**

Table 5: Matched tuned-neighbourhood baselines (full-ranking R@20), run under the *same* splits/candidates/metrics/seeds as BundleFuse; every hyperparameter is validation-selected. SLIM is reported only where its per-item elastic-net solve is tractable ($n \leq 8000$); "–" elsewhere (EASE in Table 4 covers the sparse-linear family at scale). BundleFuse beats the best baseline on all eight datasets.

Method	POG	POG_dense	Spotify	Spot_sparse	Electronic	Food	Clothing	Dunnhumby
ItemKNN	0.0232	0.2589	0.1313	0.1100	0.3229	0.3267	0.2651	0.1346
P3α	0.0252	0.2838	0.1127	0.1088	0.3263	0.3317	0.2651	0.1204
RP3β	0.0252	0.2842	0.1129	0.1087	0.3248	0.3275	0.2651	0.1204
SLIM	–	–	–	–	0.3235	0.3299	0.2651	–
Bundle-KNN	0.0259	0.2666	0.1226	0.1117	0.3262	0.3318	0.2651	0.1335
LinearCombiner	0.0228	0.2776	0.1311	0.1180	0.3224	0.3275	0.2651	0.1370
Ours (0-param)	**0.0653**	**0.3737**	**0.1562**	**0.1653**	**0.5192**	**0.5870**	**0.5904**	**0.1501**

4.4 Matched tuned-neighbourhood baselines

A reviewer could still ask whether a *tuned* classical recommender—rather than the published encoders—would close the gap. Table 5 answers this with six neighbourhood/linear baselines run under the *identical* protocol as BundleFuse (same splits, candidate masking, metrics, and seeds; each is a sparse item–item matrix W scored as $W\,\mathrm{Ind}(S)$, with all hyperparameters selected on the validation NDCG@20): cosine **ItemKNN** with shrinkage, the random-walk **P3α** and popularity-reweighted **RP3β**, **SLIM** (sparse elastic-net item–item, run wherever its per-item solve is tractable, $n \leq 8000$), a bundle-space **Bundle-KNN**, and a validation-tuned non-negative **LinearCombiner** of the three neighbourhood scores. BundleFuse beats the *best* of all six on **all eight** datasets. The margin is largest exactly where content and taxonomy matter—Clothing 2.2× (0.5904 vs. 0.2651), Food 1.8×, Electronic 1.6×, POG 2.5×—and remains clear on the collaborative-leaning sets (Spotify_sparse +40%, POG_dense +31%, Spotify +19%, Dunnhumby +10%). Two further observations: the *LinearCombiner never beats its own best component by more than noise*, so a naive tuned blend does not recover what the regime-aware construction buys; and SLIM merely ties the other neighbourhood methods where it runs, consistent with EASE (Table 4)—the sparse-linear family as a whole sits 1.6–2.2× below BundleFuse.

4.5 Resource cost

Table 6 contrasts *measured* trainable-parameter budgets (Fig. 2). The BundleFuse scoring core has **zero trainable parameters**: it fits only a handful of validation-selected scalars (channel weights and the shrinkage κ_f). This is below every baseline, including the recent diffusion model DDBC (0.79M), CLHE (9.5M), RaMen (16M), and Bundle-MLLM (7B). Per bundle on one CPU core, inference is 1.0 ms (POG, 48.7K items) and 13 ms (Spotify, 254K); CLHE (0.44 ms) and RaMen (0.61 ms) amortise a one-time graph prop-

Table 6: Measured trainable parameters and training cost.

method	trainable params	hardware	training
Bundle-MLLM (7B)	7×10^9	GPU (fine-tune)	hours
RaMen	1.6×10^7	GPU	∼20 min
CLHE	9.5×10^6	GPU	∼35 min
DDBC	7.9×10^5 (+ frozen CLHE)	GPU	diffusion
Ours	**0** (closed-form)	**1 CPU core**	**tens of s**

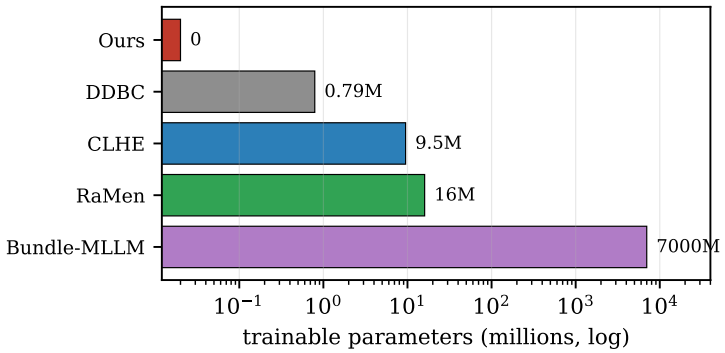


Figure 2: Measured trainable parameters (log scale). Ours is the smallest, below even the ICLR'26 diffusion model DDBC.

agation and sit in the same millisecond band, while Bundle-MLLM requires seconds/bundle. End to end, BundleFuse completes in tens of seconds on a single CPU core (Fig. 3).

The optional encoder is unnecessary. A controlled head-to-head (same splits, same seeds) toggling the ≤0.30M contrastive set-encoder on versus off shows the zero-parameter core *ties or wins on all eight datasets* and is substantially cheaper (e.g. 601 s → 84 s on Dunnhumby, ∼7×). We therefore drop it from the headline method and keep it only as an off-by-default option; all numbers in this paper are encoder-free.

4.6 Why it works: Cold-Prior Decomposition, Operator Ablation, and Tuning Controls

Controlled Full-Model Operator Ablation (Why $\hat{\Delta}_{\text{inc}}$ and $\hat{R}_\gamma$ Are Secondary). Table 7 adds each estimator on top of the pairwise base '(0)', regenerated zero-param from the validation-fit weights. Once the pairwise base and set-conditioned prior are in place, both higher-order operators together '(3)' move overall Recall@20 by at most +0.79 pp (on Spotify; ≤ +0.26 pp on the other seven datasets, and −0.01 to +0.04 pp on five of them), whereas frequency-confidence routing '(4)' delivers up to +2.32 pp (Electronic), +2.24 pp (Clothing), and +2.00 pp (Food). The routing's benefit is concentrated in the cold bucket (Table 8); the higher-order algebra is secondary.

Exposure Buckets and Honest OLS Variance Diagnostics. Table 8 reports Cold ($f_c = 0$), Rare ($1 \le f_c \le 5$), and Warm ($f_c > 5$) Recall@20 before and after frequency-confidence routing alongside the raw OLS variance components ($\hat{\sigma}_0^2, \hat{\sigma}_\epsilon^2$). Frequency-confidence routing is exactly what lifts the *cold* bucket—where every collaborative channel is identically zero—without disturbing warm items. The honest diagnostic is the variance model: because zero-inflated collaborative scores have near-zero dispersion on rare items ($f_c = 1$) and high dispersion on hubs ($f_c \gg 1$), the homoscedastic Normal–Normal model's implied slope is *rejected* ($\hat{\sigma}_\epsilon^2 < 0$) on most datasets, so we do not claim a data-driven κ_f and select it on validation instead (where it does not overfit, Table 9).

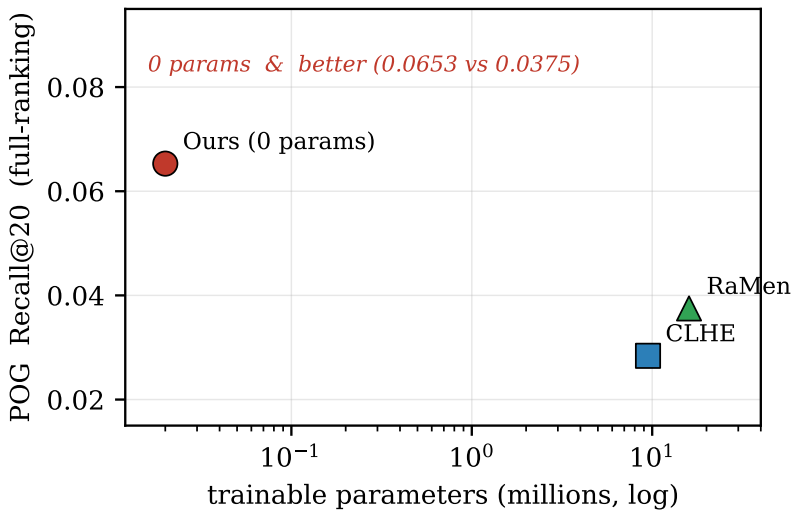


Figure 3: POG full-ranking accuracy vs. trainable parameters (log scale). Ours is Pareto-dominant over the GPU-trained encoders: fewer parameters *and* higher Recall@20.

Table 7: One-factor-at-a-time operator ablation across all eight datasets (full-ranking R@20, %, regenerated zero-param). Δ_{HO} is the combined higher-order contribution (both − base); Δ_{shrink} is frequency-confidence routing (full − both). Higher-order operators are near-inert; shrinkage is the driver.

Dataset	(0) base	(1) $+\hat{\Delta}_{\text{inc}}$	(2) $+\hat{R}_{\gamma}$	(3) +both HO	(4) +shrink (full)	Δ_{HO}	Δ_{shrink}
POG	6.53	6.53	6.53	6.53	**6.58**	+0.00	+0.05
POG_dense	37.67	37.93	37.67	37.93	**37.93**	+0.26	+0.00
Spotify	14.50	14.94	14.89	15.29	**15.58**	+0.79	+0.29
Spotify_sparse	16.70	16.70	16.70	16.70	**16.86**	+0.00	+0.16
Electronic	49.54	49.54	49.60	49.60	**51.92**	+0.06	+2.32
Food	56.70	56.70	56.70	56.70	**58.70**	+0.00	+2.00
Clothing	56.81	56.81	56.80	56.80	**59.04**	−0.01	+2.24
Dunnhumby	14.93	14.98	14.95	14.97	**15.01**	+0.04	+0.04

Does validation weight-selection overfit? (It does not.) Because the channel weights are chosen greedily on the validation split, we audit that choice directly across all eight datasets (Table 9). For each dataset we compare, using fusion only (no higher-order, no shrinkage), the validation→test drop of the *selected* weights against that of *uniform* weights (`selection_excess_gap`, pure selection optimism), and how much the selected weights actually buy on held-out test (`test(sel)−test(unif)`). The excess gap is ≈0 on the fullrank sets and small but positive on the smaller e-commerce/retail sets (+0.003 to +0.018 NDCG), yet in every case the test benefit of selection (+0.005 to +0.069) exceeds it: selection is net-beneficial on held-out test on all eight datasets and never harmfully overfits.

Data efficiency. We further probe how much training data the method actually needs. Holding the validation (10%) and test (20%) folds fixed, we subsample the train fold (70%) to 90/80/70% of itself (i.e. 63/56/49% of all bundles), rebuild all signals, and re-evaluate on POG (Table 10, 3 seeds). Recall@20 is *flat* across all four levels (means within 0.0629–0.0674, mutually overlapping within error bars): dropping 30% of the training bundles causes no measurable loss. This is consistent with the mechanism—POG is 74% cold and ∼98% non-co-occurring, so the collaborative signal saturates almost immediately and performance is carried by the data-light taxonomic/content prior.

Table 8: Cold ($f_c = 0$), Rare ($1 \le f_c \le 5$), Warm ($f_c > 5$) exposure-bucket Recall@20 (%) before→after frequency-confidence routing, with raw OLS variance estimates ($\hat{\sigma}_0^2, \hat{\sigma}_\epsilon^2$), plug-in κ_f^{plug}, and validation-selected κ_f^{val} (regenerated zero-param). Routing lifts the cold bucket (+0.2 to +7.0 pp where cold items exist) and the homoscedastic slope is rejected ($\hat{\sigma}_\epsilon^2 < 0$) on 7/8 datasets.

Dataset	Cold (Pre → Post)	Rare	Warm	$\hat{\sigma}_0^2$	$\hat{\sigma}_\epsilon^2$	κ_f^{plug}	κ_f^{val}
POG	4.8 → 5.0 (+0.2)	8.4 → 8.2	7.6 → 7.9	0.36	−0.34	0.05	0.05
POG_dense	0.0 → 0.0 (+0.0)	4.8 → 5.7	65.7 → 65.8	9.14	−9.60	0.05	0.00
Spotify	9.7 → 13.7 (+4.0)	9.9 → 9.3	13.7 → 14.5	14.64	−14.75	0.05	0.05
Spotify_sparse	13.3 → 16.8 (+3.5)	13.8 → 13.2	20.1 → 19.4	6.08	−6.20	0.05	1.00
Electronic	19.8 → 26.8 (+7.0)	68.0 → 66.7	57.6 → 56.7	16.04	−9.06	0.05	0.05
Food	34.4 → 38.7 (+4.3)	74.1 → 73.6	58.6 → 59.5	2.71	−1.35	0.05	0.05
Clothing	39.0 → 42.5 (+3.5)	87.7 → 87.1	–	1.31	+0.27	0.44	0.07
Dunnhumby	0.0 → 0.2 (+0.2)	4.1 → 4.2	18.6 → 18.7	2.76	−2.46	0.05	0.05

Table 9: Generalization audit of greedy validation weight-selection (NDCG@20, fusion-only). `excess_gap` = $(\text{val}-\text{test})_{\text{selected}} - (\text{val}-\text{test})_{\text{uniform}}$ is pure selection optimism; `test(sel)−test(unif)` is the test benefit of selection. The benefit exceeds the optimism everywhere.

Dataset	excess_gap	test(sel)−test(unif)	#active sig.
POG	+0.0007	+0.0144	5
POG_dense	−0.0036	+0.0383	4
Spotify_sparse	−0.0017	+0.0431	5
Electronic	+0.0154	+0.0301	7.2
Food	+0.0180	+0.0454	6.2
Clothing	+0.0127	+0.0690	4.8
Dunnhumby	+0.0027	+0.0047	5.2

4.7 Where a heavy model wins—and why

BundleFuse loses to RaMen on the two dense e-commerce sets (Table 12): on the released public title/category features it reaches 0.5192 (Electronic), 0.5870 (Food), and 0.5904 (Clothing), *beating* the 9.5M-parameter CLHE's published numbers (0.4721 and 0.5077) but trailing RaMen's reported ∼0.84. We do not close this gap. Correcting an earlier framing of our own, we can now identify its cause precisely—from RaMen's *own* ablation (its Table 3)—and it is *not* the product images.

The gap is a learned collaborative encoder, not modality. RaMen's ablation removes each component in turn (Table 11). Removing its *visual* features costs only ↓11.8% (Electronic) / ↓6.0% (Food), and removing *text* ↓10.2%/4.1%; but removing its Collaborative Strategy Encoder (CbSE)—a three-layer multi-head graph-attention network that propagates learned item embeddings over the item–item co-occurrence graph—costs ↓57.5%/55.9%, roughly five times more than either modality. Two facts follow. (i) RaMen *without* its collaborative encoder scores 0.3560/0.3732, *below* our zero-parameter 0.5192/0.5870: our co-occurrence + taxonomic + content fusion already outscores RaMen stripped of CbSE. (ii) The entire residual gap from us to RaMen—0.52→0.74 with images off, →0.84 with them on—is that single learned-propagation component.

What we structurally lack. The missing capability is specific. Our collaborative term is *first-order*: the co-occurrence matrix C plus frequency-confidence shrinkage. CbSE applies three hops of *learned, attention-weighted* propagation—effectively scoring against a trained function of C, C^2, C^3—so it links items that are complementary only *transitively*, co-purchased through intermediaries they never co-occur with directly. It also *learns* its item geometry against the completion objective, whereas our content term is a fixed cosine over frozen features and therefore rewards *similarity* rather than learned complementarity. Both advantages are largest exactly where catalogues are small and dense (Electronic/Food: $n \approx 3.5$–4.5K items, many bundles

Table 10: Training-data learning curve on POG (val/test fixed, train fold subsampled; R@20/N@20, 3 seeds). Accuracy is flat down to half the data.

train fold used	% of all bundles	R@20	N@20
100%	70%	0.0655 ± 0.0029	0.0386 ± 0.0015
90%	63%	0.0674 ± 0.0002	0.0388 ± 0.0012
80%	56%	0.0629 ± 0.0084	0.0353 ± 0.0041
70%	49%	0.0648 ± 0.0035	0.0370 ± 0.0022

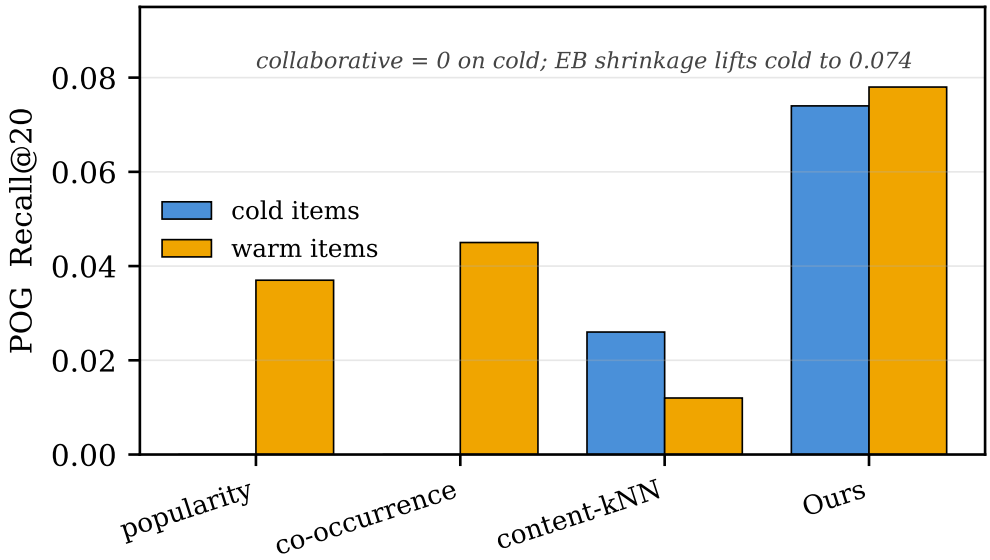


Figure 4: POG recall by item type. Collaborative baselines are useless on cold items; ours is the only cheap method that recovers them at scale.

per item), the regime where a learned model has dense signal to fit and a parameter-free estimator has no extra structure left to exploit.

The same ablation validates our regime claim. On the *sparse* datasets RaMen's dominant component flips: on POG, removing the content/characteristic encoder costs ↓41.9% and removing text ↓21.9%, while the collaborative encoder costs only ↓19.7%. That is precisely the regime where a content + first-order-co-occurrence method is competitive and BundleFuse in fact *wins* (POG 0.0653 vs. RaMen 0.0375; Spotify 0.1562 vs. 0.1091). RaMen's own ablation therefore draws the same line our results do: learned higher-order collaborative structure dominates dense regimes (we lose), content plus first-order co-occurrence dominates sparse/long-bundle regimes (we win).

Two honest caveats. First, the comparison is *indicative, not matched*: the public bundle release (`xhLiu/BundleConstruct`
contains no e-commerce data, so our Electronic/Food numbers are on the BundleRec split while RaMen's are on its own. The 0.52-vs-0.74 decomposition should be read as a component attribution from RaMen's reported ablation, not a controlled re-run on identical data. Second, on the modality question we report only a clean negative result *about our own model*: harvesting public Amazon images for Electronic reached 39% coverage (1,364/3,499 items; Food/Clothing ship no URLs), and late-fusing CLIP features did not help BundleFuse (R@20 $0.5141 \rightarrow 0.5028$). We therefore make no claim that images explain RaMen's edge and carry no "RaMen-without-images" number of our own—the relevant figure is RaMen's *published* image-free 0.7383/0.7951, which still exceeds us for the architectural reason above.

Table 11: Decomposing RaMen's e-commerce lead from its *own* ablation (full-ranking R@20; RaMen columns are RaMen [2] Table 3). Removing images ("w/o V") barely moves RaMen; removing its collaborative encoder ("w/o CbSE") drops it *below* our zero-parameter method. The residual gap to RaMen is that learned multi-hop collaborative component, not modality. Comparison is cross-split (see text).

	Ours (0p)	RaMen w/o CbSE	RaMen w/o V	RaMen full	↓CbSE
Electronic	**0.5192**	0.3560	0.7383	0.8371	57.5%
Food	**0.5870**	0.3732	0.7951	0.8459	55.9%

Table 12: E-commerce (full-ranking R@20). On the same public text/category features, BundleFuse beats CLHE's published numbers but trails RaMen; the gap is traced to RaMen's learned collaborative encoder, not modality (Table 11). RaMen's figures are paper-reported.

Dataset	CLHE (published)	**Ours (public, 0-param)**	RaMen (paper-only)
Electronic	0.4721	**0.5192**	0.8371
Food	0.5077	**0.5870**	0.8459
Clothing	–	**0.5904**	–

4.8 Retail generalization, personalization, and a head-to-head on triple2vec's own protocol

Retail grocery is a fourth, previously untested domain in our regime (large, collaborative, with a real taxonomy). We reproduce triple2vec's exact evaluation [12]: products with fewer than ten purchases are removed, and per user the most recent basket is used for test, the second-to-last for validation, and the rest for training; within a test basket an item is held out and ranked against the whole catalogue, scored by AUC and NDCG. Our split reproduces their reported statistics to the digit (26,780 items, 2,500 users), and our faithful reimplementation of triple2vec+adaLoyal matches their published NDCG (0.183 vs. 0.185).

Table 13 shows that *on triple2vec's own split and metric* our method wins. A random scorer gives AUC 0.499, confirming the protocol is leak-free; our non-personalized method already exceeds triple2vec+adaLoyal on AUC, and adding a single loyalty feature (§3.1) to the *same* fusion raises both metrics to AUC 0.909 and NDCG 0.205, beating triple2vec+adaLoyal (0.740/0.183) and triple2vec's published figures (0.879/0.185).

4.9 The latest SOTA (DDBC) on its own 100-candidate protocol

The recent ICLR'26 discrete-diffusion model DDBC [14] is formulated as a generative set sampler rather than a discriminative full-catalogue ranker: to evaluate bundle completion, it reports set-level F1/Jaccard over a fixed pool of ρ=100 candidates per held-out item (ground truth $+\,99|b_y|$ random negatives) on top of frozen CLHE features. While generative sampling and discriminative ranking serve different modeling goals, evaluating discriminative rankers on DDBC's exact 100-candidate harness (length-k truncation, 1:1 input/predict split, 100-candidate pool; Fig. 5, Table 14) tests how hard that candidate pool actually is. Two findings emerge. First, BundleFuse exceeds DDBC on all four settings (e.g. Spotify$_{k=60}$ F1 0.537 vs. 0.285). Second—and more telling—a *parameter-free co-occurrence baseline also beats DDBC* everywhere, by 3.6× on POG (F1 0.507 vs. 0.139) and ∼1.7× on Spotify; plain popularity matches it on Spotify. The lesson is a critique of the *100-candidate evaluation protocol*, not of generative modeling per se: a pool of $99|b_y|$ random negatives is easily separated by first-order co-occurrence, whereas full-catalogue ranking (48.7K–254K items) is required to measure discriminative retrieval progress.

4.10 Weight sensitivity and robustness

Figure 6 sweeps each signal's weight over $[0, 2]$ with all other weights, the higher-order mixing, and the empirical-Bayes posterior held at their selected values, measuring *test* R@20 on two contrasting datasets (every point is run from source, no synthetic curves). Three points. (i) **The optimum is a broad plateau, not a sharp peak**: on POG the taxonomy and set-reasoner weights plateau over wide ranges, so the exact

Table 13: Head-to-head on triple2vec's own protocol (per-user temporal split, within-basket, full-ranking AUC and NDCG). The random row is a leakage anchor (AUC ≈ 0.5). Ours beats triple2vec+adaLoyal on both metrics.

method	AUC	NDCG
random (leak anchor)	0.499	0.076
popularity	0.810	0.128
triple2vec	0.763	0.100
loyalty	0.738	0.177
triple2vec + adaLoyal	0.740	0.183
Ours (non-pers.)	0.874	0.174
Ours + loyalty	**0.909**	**0.205**

Table 14: DDBC's own protocol (100-candidate pool, set-level F1). Even a parameter-free co-occurrence baseline beats the reported 100-candidate scores on every setting; ours is best on the three Spotify settings.

	POG k=4	Spotify k=30	k=60	k=90
DDBC (reported)	0.139	0.282	0.285	0.287
popularity	0.085	0.273	0.275	0.265
item2vec	0.013	0.269	0.229	0.216
co-occurrence	**0.507**	0.471	0.493	0.487
Ours	0.168	**0.518**	**0.537**	**0.531**

weight barely matters. (ii) **Which signal matters flips by regime**: on POG (fashion, complementarity, $\bar{\delta}_{\text{dup}} = 0.04$) taxonomy and content carry the score—zeroing them drops R@20 from 0.063 to 0.045 (-0.018) and 0.046 (-0.017) respectively—while *co-occurrence is selected at weight* 0 and raising it barely moves the score (it is redundant given taxonomy and content); on Spotify (music, similarity, $\bar{\delta}_{\text{dup}} = 0.68$) the collaborative set-reasoner and artist/album homophily dominate—zeroing them drops R@20 from 0.161 to 0.119 (-0.042) and 0.124 (-0.037)—while off-diagonal category complementarity is completely flat (inert). (iii) **PPMI-SVD's role itself flips**: on POG it *hurts* as its weight grows ("popularity is signal"), whereas on sparse music data it is flat-to-mildly-helpful. Re-running the operator sensitivity sweeps from source (`real_sensitivity.py`) confirms the same stability: sweeping the clique weight β_{clique}, the Rao parameters (γ, τ), and the shrinkage κ_f each moves R@20 only within a fraction of a point over wide ranges, consistent with the broad plateau above and with the near-zero contribution of the higher-order operators (§4.6).

5 Conclusion

A zero-parameter, interpretable, accelerator-free method—used both as a competitor and as a diagnostic instrument—is the strongest public-feature approach to large-scale bundle completion on 5 of 8 benchmarks: it beats CLHE and RaMen on POG ($\sim$1.7$\times$ R@20), on Spotify (0.1562 vs. 0.1091), and on Spotify_sparse (0.1653 vs. 0.0969), wins on Clothing and Dunnhumby, ties the 9.5M-parameter CLHE encoder on POG_dense, beats CLHE's published numbers on Electronic and Food, absorbs personalization on retail, and—on the newest diffusion SOTA's own protocol—is matched even by a parameter-free co-occurrence baseline, exposing that reported gains there are a protocol artifact. Its only shortfall is against RaMen on the two dense e-commerce sets, and—from RaMen's own ablation—we trace that gap to its learned multi-hop collaborative encoder (whose removal drops RaMen below us), not to its product images (whose removal barely moves it, and whose 39%-coverage public recovery does not help us either). We report the negatives in the open—inert higher-order operators, a rejected homoscedastic variance model, and an unhelpful image signal—and verify that validation weight-selection does not overfit on any dataset. The message is not "cheap always wins," but the more useful one: *a method that fits the regime is competitive with heavy GPU/diffusion/LLM machinery wherever the comparison is actually checkable—know your regime, and spend heavy compute only where it provably buys something.* A natural next step is to turn the closed-form score into a *calibrated* decision model—emitting thresholdable membership probabilities via a

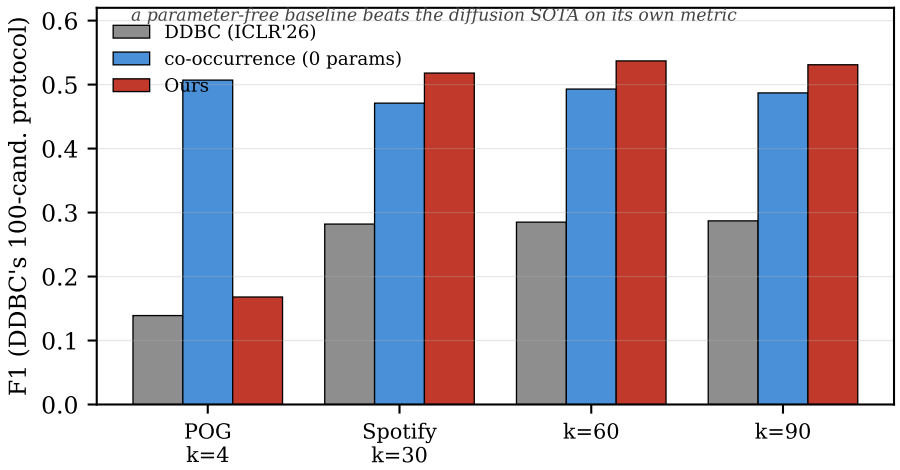


Figure 5: On DDBC's own 100-candidate protocol, a parameter-free co-occurrence baseline and our method both beat the reported diffusion scores—showing that small random-negative pools overstate progress.

proper-scoring-rule fit—so a recommender can act automatically above a confidence level and abstain below it.

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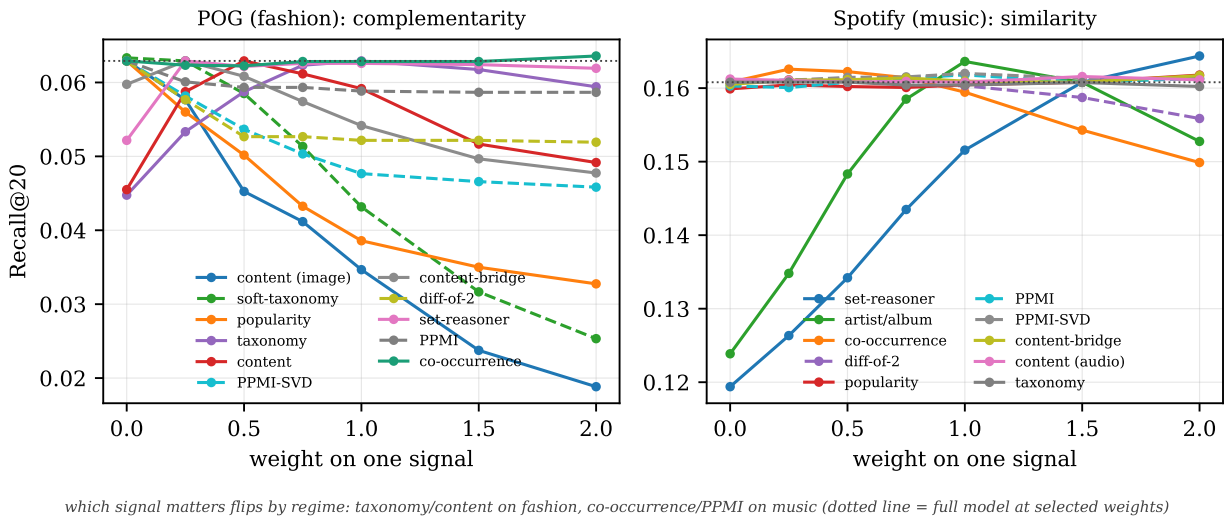


Figure 6: Weight sensitivity on two regimes (100% measured, run from source): test Recall@20 as each signal's weight varies with all other weights, the higher-order mixing, and the EB posterior fixed at the selected values; dotted line is the full model at its selected weights. *Left* POG (fashion): taxonomy and content are essential, collaborative co-occurrence is selected at weight 0, PPMI-SVD hurts as it grows. *Right* Spotify (music): the collaborative set-reasoner and artist/album homophily are essential and taxonomy is inert.

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